Personal Server

Current state and general plan for Leandro's server

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By Leandro Chamorro

Contents

1 Objective **2**

2 Project scope **2**

3 General plan **3**

4 Current state **4**

4.1 Hardware and storage 4

5 Final product acceptance criteria **4**

6 Future work **4**

7 References **4**

1 Objective

This document describes the current state and general plan of my personal server. I'm currently using an old laptop to store personal files, provide private remote access and run a separate environment for learning and testing. It is also an opportunity to practice Linux administration, networking, virtualization, databases and Python deployment.

The goal is to access my files and applications from my authorized devices while keeping personal services separate from experimental work. The server will be built in stages, with a specific result to verify before moving to the next one.

2 Project scope

- Provide secure and restricted access to two authorized PCs.
- Keep the laboratory isolated from the home network, the administrative services of the host and the services VM.
- Use different storage units for the host, VM system disks and databases and other big files saved.
- Collect lightweight host measurements, maintain logs and test alerts and recovery procedures.

3 General plan

The following plan divides the work into work blocks, each one adding functions to the systetm. Stages 7B and 8B are later extensions and can be developed independently, the server should have the main functions available in block 6.

Work block	Description	Acceptance criteria
1 Prepare administration and networking	Configure Ubuntu, SSH keys, UFW, automatic security updates, configuration history and local recovery.	Key-based access works after a restart, UFW is active, local recovery works and unauthorized LAN devices cannot reach SSH.
2 Establish private remote access	Configure Tailscale, restrictive Grants and separate SSH keys for both PCs. Protect device enrollment with Tailnet Lock.	A new SSH session works from another network; unauthorized access is denied and recovery remains available.
3 Prepare virtualization and isolation	Validate the HDD connection, make the VM firewall persistent and create separate NAT networks and VMs. Allocate resources from measurements.	Isolation passes positive and negative IPv4/IPv6 checks and survives restart. The services VM cannot start without its expected data mount.
4 Install file services	Install Nextcloud and PostgreSQL in the services VM, with files on the HDD-backed data disk and the database on the SSD.	Services restart correctly using test data. A database dump restores into a separate temporary database.
5 Configure private HTTPS and accounts	Set up private HTTPS, separate administrator and daily-use accounts, two-factor authentication, permissions and file versions.	Authorized devices can use the portal; account separation and file-version recovery work, with public links disabled.
6 Verify recovery, alerts and host monitoring	Check access rules, local recovery, logs, lightweight metrics and scheduled collection with bounded retention.	Only intended routes work; a test alert arrives and collection resumes after restart within its storage limits.
7A Verify the minimum backup	Have the PC pull consistent files, database dumps, configuration and logs into encrypted, versioned storage outside the server.	Files and a database are restored successfully. Personal data are introduced only after this stage is accepted.
7B Extend backup protection	Add a separate offline copy and document rotation, retention and complete recovery.	The full service can be recovered in a test environment from an independent offline copy.
8A Deploy the practice laboratory	Run a simulated equipment dashboard and public-data nowcasting in an isolated VM, using reproducible Python environments and service automation.	Private access, isolation, restart recovery, processing tests and reproducibility are verified. Technical results are measured and recorded.
8B Evaluate later extensions	Consider extra applications, tools or data sources when there is a concrete need and enough resources.	Each addition has a defined purpose, maintenance procedure and verified access rules.

4 Current state

The host administration and private-access configuration have been implemented. Stage 2 has also been completed. Stage 3 is in progress, virtualization packages are installed, but virtual machines have not been created yet since a problem with the external hard drive arose and is currently being solved.

4.1 Hardware and storage

The server is an HP Pavilion notebook with an Intel Core i5-7200U and four hardware threads. Ubuntu reports approximately 11 GiB of usable RAM, but there are 12 GB installed. The battery is not functional and the display backlight is damaged, so the notebook depends on its charger and can be managed locally with an external monitor and keyboard.

For storage, the server has a SATA SSD that provides 223.6 GiB of visible capacity and contains the host system. The external hard drive is a Western Digital HDD and has 500 GB nominal capacity, or 465.8 GiB visible, and connects through a USB bridge. The usb that connects the HDD to the server was moved by mistake and disconnected the device, so evaluations on the drive's status are being conducted.

The intended distribution places VM system disks and PostgreSQL on the SSD, with a dedicated services-VM data disk on the HDD. Space must remain available for file growth, Nextcloud versions and deleted-file retention.

5 Final product acceptance criteria

- Authorized PCs can administer the host privately, local recovery works and unauthorized devices cannot reach administrative services.
- The services VM and laboratory VM have separate networks and credentials, with isolation demonstrated by connection tests after a restart.
- Nextcloud is accessible through private HTTPS with two-factor authentication, verified account permissions and working file recovery.
- The HDD connection is stable under the defined workload, and the services VM cannot silently use an unintended directory when its data mount is absent.
- An encrypted, versioned backup outside the server restores files and the database before personal data are stored on the server.
- Host collection and alerts survive restart, respect retention limits and produce a verified test notification.
- The laboratory can be rebuilt from code, pinned dependencies with hashes and experiment manifests preserved on the PC.

6 Future work

After the main plan is done, my objective is to use the laboratory to develop personal projects or applications. If possible, I would like to add new devices to the network like personal portable devices or allow access to known people devices. The later would require an extreme security verification. Physical sensors, additional experimental vibration datasets and larger monitoring tools remain possible extensions. Their adoption will depend on a concrete learning objective, available resources and any changes required to the access rules. Other physical updates to the server environment is under consideration, but will be evaluated once the project is fully functional.

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